BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Selection: IF Statements

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. When is A AND B true?

2. What does CPU cache hold?

A) Only saved personal files
B) Only website passwords
C) Often-used data and instructions
D) Only the next address

3. "A register is a large external drive used for backups." Correct this. Explain why.

4. The next instruction is at address 240. An addition gives 18. What do the accumulator and Program Counter hold? Explain the difference.

Today's objectives

Use IF statements to make decisions. Use selection and if in a practical scenario.

Key vocabulary - Read, cover, recall

Selection

Selection lets a program follow different instructions according to a condition.

IF

An IF block executes its instructions only when the condition is true.

Comparison

The Boolean test must compare values rather than assign a new value.

ENDIF

ENDIF marks the end of the IF statement so later instructions can run normally.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

What is Selection?	Selection - Selection lets a program follow different instructions according to a condition.
In this example	A theme-park ride height rule uses if; An IF block executes its instructions only when the condition is true.
Follow the effect	The Boolean test must compare values rather than assign a new value. Explain how this changes the eventual program result.
Check understanding	ENDIF is used for a specific reason: ENDIF marks the end of the IF statement so later instructions can run normally.

I Do - One worked example

A theme-park ride height rule is being written. Explain how selection and if are used to make the program work. [4 marks]

1. We are designing a theme-park ride height rule. Identify which data and decisions are essential.

2. Selection [1] - Selection lets a program follow different instructions according to a condition. [1].

3. IF [1] - An IF block executes its instructions only when the condition is true. [1].

4. Two linked points for each technique give four mark points.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a school competition entry, explain how selection helps the program. [2 marks]

2. For a sports pass check, explain why if matters. [2 marks]

3. When designing a exam pass notification, explain how comparison and endif should be used. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the IF condition that switches on ventilation above 30 degrees.

```
temperature = 32
if ... then
print("Fan on")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. State two programming terms or techniques relevant to this lesson. [2 marks]

Q3. A cinema age check needs to work correctly. Explain how if and comparison would help. [4 marks]

Q4. A pupil is programming a library opening status. Explain the roles of selection, comparison and endif in their solution. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Selection lets a program follow different instructions according to a condition. ☐ T ☐ F
2. Selection - Selection lets a program follow different instructions according to a condition. ☐ T ☐ F
3. A theme-park ride height rule uses if; An IF block executes its instructions only when the condition is ... ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Only when both A and B are true. ☐

Q2 C - Often-used data and instructions ☐

Q3 A small, fast storage location inside the CPU. It holds data or addresses in use. ☐

Q4 The accumulator holds the result, 18. The PC holds the next instruction's address, 240. ☐

Code First - Model

```
temperature = 32
if temperature > 30 then
print("Fan on")
endif
```

Output: Fan on

We Do - Check the model

1. Selection [1] - Selection lets a program follow different instructions according to a condition. [1]. ☐

2. IF [1] - An IF block executes its instructions only when the condition is true. [1]. ☐

3. Comparison [1] - The Boolean test must compare values rather than assign a new value. [1]. ENDIF [1] - ENDIF marks the end of the IF statement so later instructions can run normally. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Selection [1]; Comparison [1]. ☐

Q3 IF [1] - An IF block executes its instructions only when the condition is true. [1]. Comparison [1] - The Boolean test must compare values rather than assign a new value. [1]. ☐

Q4 Selection [1] - Selection lets a program follow different instructions according to a condition. [1]. Comparison [1] - The Boolean test must compare values rather than assign a new value. [1]. ENDIF [1] - ENDIF marks the end of the IF statement so later instructions can run normally. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: